

LESIBA JAMES KGANYAGO

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PROFILE

Data Scientist with hands-on experience building end-to-end machine learning projects and extracting actionable insights from data. Skilled in Python, SQL, and data visualization, with practical experience in predictive modeling, time series forecasting, and business-focused analytics. Proven ability to apply data science techniques to real-world problems through projects and consulting work. Currently expanding knowledge in advanced machine learning and production-level deployment.

PROFESSIONAL & CONSULTING EXPERIENCE

Studio Intelligence AI – Data Science Consultant | 2024–2025

- Led end-to-end development of ML platform using Databricks and Python to optimize music studio revenue and operations
- Developed three predictive models:
 - ✓ Revenue forecasting model (regression) improving pricing strategy accuracy by 8%, enabling data-driven rate adjustments across peak/off-peak hours
 - ✓ Booking demand time-series model reducing scheduling gaps by 15% increasing studio utilization
 - ✓ Artist retention classifier identifying high-risk churn clients with 84% precision, enabling targeted retention strategies
- Built Power BI dashboards translating model outputs into pricing and scheduling decisions

Local Butcher Shop – Data Analytics Consultant | 2024

- Designed and executed A/B testing framework that identified product bundling increased conversions by 15%
- Analyzed sales data using SQL and Python, providing data-driven recommendations for inventory management
- Technologies: Python, SQL, Statistical Testing, Data Visualization

DATA SCIENCE PROJECT PORTFOLIO

UbuntuCare – Wait Time Prediction System

- Developed XGBoost model forecasting patient wait times: $R^2 = 0.61$, MAE = 134 minutes(target range: 1–1088 min, mean: 348 min)
- Cost-benefit analysis framework: Calculated R54.1M net benefit opportunity (underestimation cost: R8M, overestimation: R435M, correct predictions: R66.5M benefit)
- Analyzed patient flow data to identify key drivers including demographics, facility capacity, and clinical risk factors
- Risk categorization: Achieved 66% precision identifying high-risk patients requiring urgent intervention

Intelligent Fraud Detection in Banking

- Built a real-time fraud detection model that reduced false positives, improved customer transaction approval rates, and minimized revenue loss from fraudulent activities.
- Actionable risk segmentation: Identified highest-risk segments (clothing retailers: 5.20% fraud, young adults 18-25: 5.24% fraud, dashboard cards: 5.59% fraud)
- Enabled targeted verification flows and device-based security tiers

TECHNICAL SKILLS

- Programming & Databases:** Python (Pandas, NumPy, Scikit-learn) | SQL (MySQL, PostgreSQL) | PySpark (Databricks)
- Data Analysis & Visualization:** Pandas | NumPy | Matplotlib | Seaborn | Plotly | Power BI | Dashboard Design | Data Storytelling
- Machine Learning & Modeling:** Machine Learning: Regression Analysis | Classification Models | Time Series Forecasting | XGBoost | Feature Engineering | Model Evaluation (SHAP, MLflow)
- Data Visualization & BI:** Power BI | Matplotlib | Seaborn | Plotly | Dashboard Design | Data Storytelling
- Cloud & Big Data:** Databricks Lakehouse | PySpark

ANALYTICAL & BUSINESS SKILLS

- Statistical Analysis:** A/B Testing | Hypothesis Testing | Statistical Inference | Experimental Design
- Business Acumen:** ROI Analysis | KPI Development | Stakeholder Communication | Business Case Development
- Project Management:** End-to-end ML Project Lifecycle | Cross-functional Collaboration | Requirements Gathering

EDUCATION & CERTIFICATIONS

BSc in Informatics | University of South Africa

- Graduated: 2024 | Relevant Coursework: Data Structures, Statistics, Database Systems, Programming

IBM Data Science Professional Certificate | 2024

- Python Project for Data Science, Data Analysis with Python, Data Visualization Machine Learning with Python
- Machine Learning with Python